

Woo-Jin CHO KIM

PERSONAL DATA

EMAIL: woojinchokimm@hotmail.com
HOMEPAGE: woojinchokimm.github.io
LINKEDIN: [woojinchokim](https://www.linkedin.com/in/woojinchokim)

WORK EXPERIENCE

Current JAN 2024	Deep Learning Scientist at Ultromics Ltd , Oxford, UK <i>AI and Computer Vision Group</i> - Developed end-to-end echo view classification models for images and videos using CNNs, RNNs, and vision-transformers. - Pioneered RL-driven strategy to reduce patient echo scan needs by up to 70% for cardiac amyloidosis detection (MICCAI 2025 submission). - Fine-tuned 🤖 foundation models for downstream tasks with LoRA and ExPLoRA. - Deployed production-ready models with Docker + AWS for scalable inference.
JAN 2024 APR 2022	Junior Deep Learning Scientist at Ultromics Ltd , Oxford, UK - Processed and managed high-volume DICOM data pipelines. - A/B tested amyloidosis detection models for FDA-regulatory compliance. - Led cross-domain validation with clinicians to validate model attention.
JAN 2022 SEPT 2021	Research Intern at Ultromics Ltd , Oxford, UK Uncertainty estimation and DICOM image pre-processing techniques.
DEC 2023 DEC 2022	PhD Candidate, KCL x Ultromics Ltd , London, UK - Innovated a novel LV volume quantification technique for echo analysis. - Now integrated into Ultromics product.

SELECTED PUBLICATIONS

- Automated Detection of Apical Foreshortening in Echocardiography Using Statistical Shape Modelling**, Cho Kim W. J.*, Leeson P., Lamata P., et al. *UMB 2023*
- Beyond Simpson's Rule: Accounting for Orientation and Ellipticity Assumptions** Cho Kim W. J.*, Lewandowski A. J., King A. P., Leeson P., Lamata P., et al. *UMB 2022*
- BackMix: Mitigating Shortcut Learning in Echocardiography with Minimal Supervision** Bransby K., Beqiri A., Cho Kim W.J., Chartsias A., Gomez A., et al. *MICCAI 2024*
- Multi-Site Class-Incremental Learning with Weighted Experts in Echocardiography** Bransby K., Beqiri A., Cho Kim W.J., Chartsias A., Gomez A., et al. *ASMUS 2024*
- Left Ventricle Contouring of Apical Three-Chamber Views on 2D Echocardiography** Gomez A., Porumb M., Judge T., Cho Kim W.J., Chartsias A., et al. *ASMUS 2022*
- Asymmetric Contour Uncertainty Estimation for Medical Image Segmentation** Judge T., Bernard O., Cho Kim W.J., Gomez A., Jodoin P. M., et al. *MICCAI 2023*

EDUCATION

AUG 2023	PhD in AI for Medical Image Computing
SEPT 2017	King's College London Dr. Pablo Lamata , Dr. Andrew King Thesis topic: Improving echocardiographic diagnostic accuracy.
SEPT 2017	MSc in AI and Machine Learning (Computing)
SEPT 2016	Imperial College London Dr. Panos Parpas
SEPT 2015	BEng in Biomedical Engineering
SEPT 2012	King's College London Dr. Oleg Aslanidi Rosalind Franklin Prize for Respiratory Motion Modelling Project, MIUA 2015

SKILLS

Tensorflow, Pytorch, Python, Jupyter, AWS, Docker, Git, SQL, Bash, OpenCV